CHAPTER 2

Electrochemistry

Electrochemical and Galvanic Cells

1. The **correct** value of cell potential in volt for the reaction that occurs when the following two half cells are connected, is *(2023-Manipur)*

$Fe^{2+}_{(aq)} + 2e^- \rightarrow Fe(s)$, $E° = –0.44$ V

$Cr_2O^{2-}_{7(aq)} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$, $E° = +1.33$ V

a. +1.77 V
b. +2.65 V
c. +0.01 V
d. +0.89 V

2. Two half cell reactions are given below. *(2022 Re)*

$Co^{3+} + e^- \rightarrow Co^{2+}$ $\quad E°_{Co^{2+}/Co^{3+}} = –1.81$ V

$2Al^{3+} + 6e^- \rightarrow 2Al(s)$ $\quad E°_{Al/Al^{3+}} = +1.66$ V

The standard EMF of a cell with feasible redox reaction will be:

a. –3.47 V
b. +7.09 V
c. +0.15 V
d. +3.47 V

3. Given below are half cell reactions: *(2022)*

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

$E^o_{Mn^{2+}/MnO_4^-} = -1.510\,V$

$\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$

$E_{O_2/H_2O} = +1.223$ V

Will the permanganate ion, MnO^-_4 liberate O_2 from water in the presence of an acid?

a. No because $E^o_{cell} = -2.733\,V$
b. Yes, because $E^o_{cell} = +0.287\,V$
c. No, because $E^o_{cell} = -0.287\,V$
d. Yes, because $E^o_{cell} = +2.733$ V

4. Consider the change in oxidation state of Bromine corresponding to different emf values as shown in the diagram below : *(2018)*

$$BrO_4^- \xrightarrow{1.82\ V} BrO_3^- \xrightarrow{1.5\ V} HBrO$$
$$HBrO \xrightarrow{1.595\ V} Br_2 \xrightarrow{1.0652\ V} Br^-$$

Then the species undergoing disproportionation is

a. BrO_3^-
b. BrO_4^-
c. HBrO
d. Br_2

5. A button cell used in watches functions as following

$Zn(s) + Ag_2O(s) + H_2O(l) \rightleftharpoons 2Ag(s) + Zn^{2+}(aq) + 2OH^-(aq)$. If half cell potentials are

$Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$; $E^o = –0.76$ V

$Ag_2O(s) + H_2O(l) + 2e^- \rightarrow 2Ag(s) + 2OH^-(aq)$, $E^o = 0.34$ V.

The cell potential will be: *(2013)*

a. 1.10 V
b. 0.42 V
c. 0.84 V
d. 1.34 V

Nernst Equation

6. The standard cell potential of the following cell $Zn|Zn^{2+}(aq)|Fe^{2+}(aq)|Fe$ is 0.32 V. Calculate the standard Gibbs energy change for the reaction : *(2024 Re)*

$Zn(s) + Fe^{2+}(aq) \rightarrow Zn^{2+}(aq) + Fe(s)$

(Given: 1 F = 96487 C)

a. $–61.75$ kJ mol$^{–1}$
b. $+5.006$ kJ mol$^{–1}$
c. $–5.006$ kJ mol$^{–1}$
d. $+61.75$ kJ mol$^{–1}$

7. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R: *(2023)*

Assertion A: In equation $\Delta_rG = –nFE_{cell}$, value of Δ_rG depends on n.

Reasons R: E_{cell} is an intensive property and Δ_rG is an extensive property.

In the light of the above statements, choose the correct answer from the options given below:

a. A is false but R is true.
b. Both A and R are true and R is the correct explanation of A.
c. Both A and R are true but R is NOT the correct explanation of A.
d. A is true but R is false

8. The $E^\ominus$ values for *(2023-Manipur)*

$Al^+/Al = +0.55$ V and $Tl^+/Tl = –0.34$ V

$Al^{3+}/Al = –1.66$ V and $Tl^{3+}/Tl = +1.26$V

Identify the incorrect statement

a. Al is more electropositive than Tl
b. Tl^{3+} is a good reducing agent than Tl^{1+}
c. Al^+ is unstable in solution
d. Tl can be easily oxidised to Tl^+ then Tl^{3+}

9. Standard electrode potential for the cell with cell reaction

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

is 1.1 V. Calculate the standard Gibbs energy change for the cell reaction.

(Given F = 96487 C mol^{-1}) *(2022 Re)*

a. –200.27 J mol^{-1} b. –200.27 kJ mol^{-1}
c. –212.27 kJ mol^{-1} d. –212.27 J mol^{-1}

10. At 298 K, the standard electrode potentials of Cu^{2+}/Cu, Zn^{2+}/Zn, Fe^{2+}/Fe and Ag^{+}/Ag are 0.34 V, – 0.76 V, – 0.44 V and 0.80 V, respectively.

On the basis of standard electrode potential, predict which of the following reaction can not occur? *(2022)*

a. $2CuSO_4(aq) + 2Ag(s) \rightarrow 2Cu(s) + Ag_2SO_4(aq)$
b. $CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s)$
c. $CuSO_4(aq) + Fe(s) \rightarrow FeSO_4(aq) + Cu(s)$
d. $FeSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Fe(s)$

11. Find the emf of the cell in which the following reaction takes place at 298 K *(2022)*

$Ni(s) + 2Ag^{+}(0.001\ M) \rightarrow Ni^{2+}(0.001\ M) + 2Ag(s)$

(Given that $E^\circ_{cell} = 10.5$ V, $\frac{2.303\ RT}{F} = 0.059$ at 298 K)

a. 1.05 V b. 1.0385 V c. 1.385 V d. 0.9615 V

12. Identify the reaction from following having top position in EMF series (Std. red. potential) according to their electrode potential at 298 K. *(2020-Covid)*

a. $Fe^{2+} + 2e^- \rightarrow Fe(s)$ b. $Au^{3+} + 3e^- \rightarrow Au(s)$
c. $K^+ + 1e^- \rightarrow K(s)$ d. $Mg^{2+} + 2e^- \rightarrow Mg(s)$

13. For a cell involving one electron E°cell = 0.59 V at 298 K, the equilibrium constant for the cell reaction is:

$\left[\text{Given that } \frac{2.303RT}{F} = 0.059V \text{ at } T = 298K\right]$ *(2019)*

a. 1.0×10^{2} b. 1.0×10^{5} c. 1.0×10^{10} d. 1.0×10^{30}

14. For the cell reaction

$2Fe^{3+}(aq) + 2I^{-}(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$

$E^{\ominus}_{cell} = 0.24$ V at 298 K. The standard Gibbs energy ($\Delta_r G^{\ominus}$) of the cell reaction is:

[Given that Faraday constant F = 96500 C mol^{-1}] *(2019)*

a. –46.32 kJ mol^{-1} b. –23.16 kJ mol^{-1}
c. 46.32 kJ mol^{-1} d. 23.16 kJ mol^{-1}

15. If the E°cell for a given reaction has a negative value, which of the following gives the correct relationships for the values of ΔG° and K_{eq}? *(2016 - II)*

a. $\Delta G^0 < 0; K_{eq} > 1$ b. $\Delta G^0 < 0; K_{eq} < 1$
c. $\Delta G^0 > 0; K_{eq} < 1$ d. $\Delta G^0 > 0; K_{eq} > 1$

16. The pressure of H_2 required to make the potential of H_2-electrode zero in pure water at 298 K is: *(2016 - I)*

a. 10^{-4} atm b. 10^{-14} atm c. 10^{-12} atm d. 10^{-10} atm

17. The pair of compounds that can exist together is: *(2014)*

a. $HgCl_2, SnCl_2$ b. $FeCl_2, SnCl_2$
c. $FeCl_3$, KI d. $FeCl_3, SnCl_2$

18. A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of pH = 10 and by passing hydrogen gas around the platinum wire at one atm pressure. The oxidation potential of electrode would be? *(2013)*

a. 0.059 V b. 0.59 V c. 0.118 V d. 1.18 V

Conductance of Electrolytic Solutions

19. The conductivity of centimolar solution of KCl at 25°C is 0.0210 ohm^{-1} cm^{-1} and the resistance of the cell containing the solution at 25 °C is 60 ohm. The value of cell constant is: *(2023)*

a. 3.34 cm^{-1} b. 1.34 cm^{-1} c. 3.28 cm^{-1} d. 1.26 cm^{-1}

20. Molar conductance of an electrolyte increase with dilution according to the equation: *(2023-Manipur)*

$\Lambda_m = \Lambda_m^\circ - A\sqrt{c}$

Which of the following statements are **true**?

A. This equation applies to both strong and weak electrolytes.
B. Value of the constant A depends upon the nature of the solvent.
C. Value of constant A is same for both $BaCl_2$ and $MgSO_4$
D. Value of constant A is same for both $BaCl_2$ and $Mg(OH)_2$.

Choose the **most appropriate** answer from the options given below:

a. (A) and (B) only b. (A), (B) and (C) only
c. (B) and (C) only d. (B) and (D) only

21. The molar conductance of NaCl, HCl and CH_3COONa at infinite dilution are 126.45, 426.16 and 91.0 S cm^2 mol^{-1} respectively. The molar conductance of CH_3COOH at infinite dilution is. Choose the right option for your answer. *(2021)*

a. 390.71 S cm^2 mol^{-1} b. 698.28 S cm^2 mol^{-1}
c. 540.48 S cm^2 mol^{-1} d. 201.28 S cm^2 mol^{-1}

22. The molar conductivity of 0.007 M acetic acid is 20 S cm^2 mol^{-1}. What is the dissociation constant of acetic acid? Choose the correct option. *(2021)*

$$\begin{bmatrix} \Lambda^\circ_{H^+} = 350\ S\ cm^2 mol^{-1} \\ \Lambda^\circ_{CH_3COO^-} = 50\ S\ cm^2 mol^{-1} \end{bmatrix}$$

a. 2.50×10^{-4} mol L^{-1} b. 1.75×10^{-5} mol L^{-1}
c. 2.50×10^{-5} mol L^{-1} d. 1.75×10^{-4} mol L^{-1}

23. The molar conductivity of a 0.5 mol dm^{-3} solution of $AgNO_3$ with electrolytic conductivity of 5.76×10^{-3} S cm^{-1} at 298 K is: *(2016 - II)*

a. 0.086 S cm^2 mol^{-1} b. 28.8 S cm^2 mol^{-1}
c. 2.88 S cm^2 mol^{-1} d. 11.52 S cm^2 mol^{-1}

24. At 25°C, molar conductance of 0.1 molar aqueous solution of ammonium hydroxide is 9.54 ohm^{-1} $cm^2 mol^{-1}$ and at infinite dilution its molar conductance is 238 ohm^{-1} $cm^2 mol^{-1}$. The degree of ionisation of ammonium hydroxide at the same concentration and temperature is: *(2013)*

a. 2.080 % b. 20.800 % c. 4.008 % d. 40.800 %

Electrolytic Cells and Electrolysis

25. Match List I with List II. *(2024)*

List I (Conversion)		List II (Number of Faraday required)	
A.	1 mol of H_2O to O_2	I.	3F
B.	1 mol of MnO_4^- to Mn^{2+}	II.	2F
C.	1.5 mol of Ca from molten $CaCl_2$	III.	1F
D.	1 mol of FeO to Fe_2O_3	IV.	5F

Choose the correct answer from the options given below:

a. A-II, B-III, C-I, D-IV　b. A-III, B-IV, C-II, D-I
c. A-II, B-IV, C-I, D-III　d. A-III, B-IV, C-I, D-II

26. Mass in grams of copper deposited by passing 9.6487 A current through a voltmeter containing copper sulphate solution for 100 seconds is: *(2024)*
(Given : Molar mass of Cu: 63 g mol^{-1}, 1 F = 96487 C)

a. 31.5 g　b. 0.0315 g　c. 3.15 g　d. 0.315 g

27. On electrolysis of dil sulphuric acid using Platinum (Pt) electrode, the product obtained at anode will be: *(2020)*

a. Oxygen gas　b. H_2S gas
c. SO_2 gas　d. Hydrogen gas

28. The number of Faradays (F) required to produce 20 g of calcium from molten $CaCl_2$ (Atomic mass of Ca = 40 g mol^{-1}) is: *(2020)*

a. 2　b. 3　c. 4　d. 1

29. During the electrolysis of molten sodium chloride, the time required to produce 0.10 mol of chlorine gas using a current of 3 amperes is: *(2016 - II)*

a. 220 minutes　b. 330 minutes
c. 55 minutes　d. 110 minutes

30. The number of electrons delivered at the cathode during electrolysis by a current of 1 ampere in 60 seconds is: (charge on electron = 1.60×10^{-19} C) *(2016 - II)*

a. 3.75×10^{20}　b. 7.48×10^{23}
c. 6×10^{23}　d. 6×10^{20}

31. When 0.1 mol MnO_4^{2-} is oxidised, the quantity of electricity required to completely oxidise MnO_4^{2-} to MnO_4^- is: *(2014)*

a. 96500 C　b. 2 × 96500 C
c. 9650 C　d. 96.50 C

32. The weight of silver (atomic weight = 108) displaced by a quantity of electricity which displaces 5600 mL of O_2 at STP will be: *(2014)*

a. 10.8 g　b. 54.0 g　c. 108.0 g　d. 5.4 g

Batteries, Fuel Cells and Corrosion

33. From the following select the one which is not an example of corrosion. *(2024 Re)*

a. Rusting of iron object
b. Production of hydrogen by electrolysis of water
c. Tarnishing of silver
d. Development of green coating on copper and bronze ornaments

34. In a typical fuel cell, the reactants (R) and product (P) are *(2020-Covid)*

a. $R = H_{2(g)}, O_{2(g)}; P = H_2O_{(\ell)}$
b. $R = H_{2(g)}, O_{2(g)}, Cl_{2(g)}; P = HClO_{4(aq)}$
c. $R = H_{2(g)}, N_{2(g)}; P = NH_{3(aq)}$
d. $R = H_{2(g)}, O_{2(g)}; P = H_2O_{2(\ell)}$

35. Zinc can be coated on iron to produce galvanized iron but the reverse is not possible. It is because: *(2016 - II)*

a. Zinc has lower negative electrode potential than iron
b. Zinc has higher negative electrode potential than iron
c. Zinc is lighter than iron
d. Zinc has lower melting point than iron

36. A device that converts energy of combustion of fuels like hydrogen and methane, directly into electrical energy is known as: *(2015)*

a. Electrolytic cell　b. Dynamo
c. Ni-Cd cell　d. Fuel cell

Answer Key

1. (a)	**2.** (d)	**3.** (b)	**4.** (c)	**5.** (a)	**6.** (a)	**7.** (b)	**8.** (b)	**9.** (c)	**10.** (a)
11. (none)	**12.** (b)	**13.** (c)	**14.** (a)	**15.** (c)	**16.** (b)	**17.** (b)	**18.** (b)	**19.** (d)	**20.** (d)
21. (a)	**22.** (b)	**23.** (d)	**24.** (c)	**25.** (c)	**26.** (d)	**27.** (a)	**28.** (d)	**29.** (d)	**30.** (a)
31. (c)	**32.** (c)	**33.** (b)	**34.** (a)	**35.** (b)	**36.** (d)				